

Integration Testing



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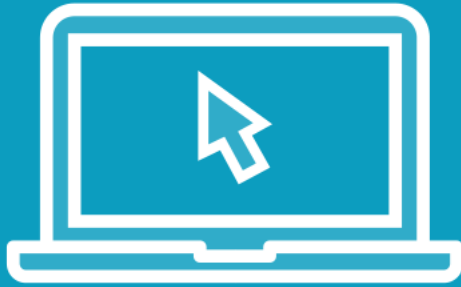
Module Overview



Integration tests using `createTestClient``
Snapshot tests and their benefits



Demo



Unit test Query resolvers

Unit test Mutation resolvers

Implement mocks in the ApolloServer
MockList

Build a schema using Introspection



Snapshot Test

Comparing the output of a function's execution with future test runs

Test Run 1 (snapshot created)

```
buildJson();  
// outputs:  
{  
  "name": "Hello",  
  "description": "world"  
}
```

Test Run 2

```
buildJson();  
// outputs:  
{  
  "name": "Hello",  
  "description": "",  
  "date": "2020-01-01"  
}
```

// test FAILS since snapshots no longer match

Why Use Snapshot Tests?

Pros

Snapshot tests are very fast
Provide a level of confidence in your code

Cons

Tedious to update snapshots when the output is supposed to change



Summary



Apollo Client Hooks of Interest

useQuery

for fetching data

useMutation

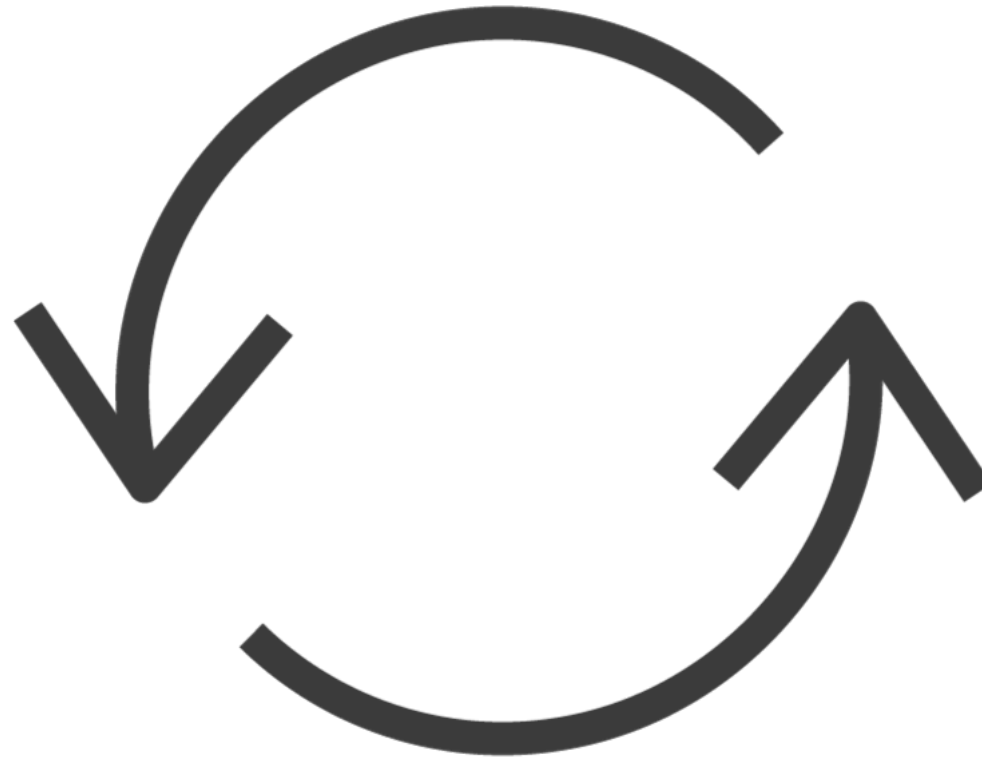
for editing data



GraphQL Database



Client Code



Resolvers

Responsible for populating data for every field in the schema

resolvers.js

```
const resolvers = {  
  Query: {  
    products: (parent, args, ctx) => {...}  
  },  
  Mutation: {  
    addProduct: (parent, args, ctx) => {...}  
  },  
  Product : {  
    name: (parent, args, ctx) => {...}  
  }  
};
```

```
const mocks = {  
  Query: () => ({  
    products: () => new MockList(5),  
    users: () => new MockList([5, 15]),  
  })  
};
```

- ◀ Return 5 mocked *products*
- ◀ Return somewhere between 5 and 15 mocked *users*



Introspection

A method used to ask a GraphQL schema for information about itself, like what queries it supports.



Testing Topics



Queries and Mutations in React



Handling errors



Unit testing resolvers



Server-side integration tests



Thank you

